

Title: Explainable Vision-Language Diagnostic Assistant for Chest Radiographs with Offline Rural Deployment

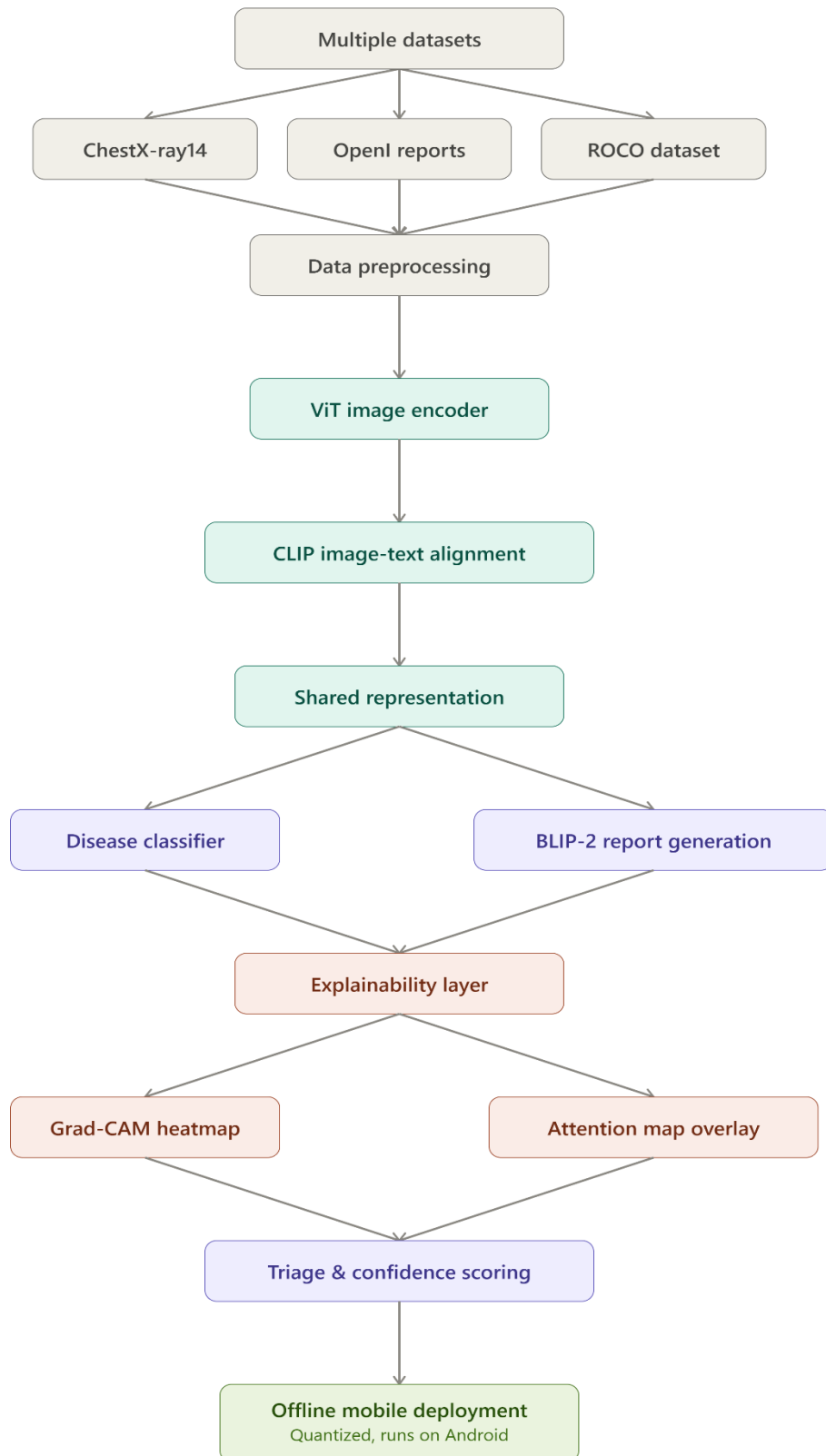
Abstract

In many rural areas, clinics don't have a radiologist — the specialist doctor who knows how to read X-ray images and spot problems like pneumonia, tuberculosis, or other lung diseases. Because of this, patients often have to travel far, wait a long time, or go without a proper diagnosis. This project builds a smart assistant that looks at a chest X-ray, just like a doctor would, and explains what it sees in simple terms — for example, "this looks like early signs of pneumonia" — along with how confident it is in that answer. It also tells the clinic staff how urgent the case is, like "please see a doctor within 24 hours," so serious cases get attention first. The system uses Vision Transformers (ViT) to understand the image, CLIP to connect the image with medical text, and BLIP-2 to generate a readable diagnosis from what it sees. Unlike a normal AI tool that just gives an answer without explaining itself, this assistant highlights the exact part of the X-ray it focused on — like circling the area of concern — so a nurse or health worker can actually see the reasoning behind the result, instead of blindly trusting the machine. The tool is also made small and lightweight enough to run directly on a basic smartphone, without needing the internet, since many rural clinics don't have reliable connectivity. This means it can work fully offline, right where it's needed most. The model is trained and tested using accuracy, precision, recall, and F1-score, and its explanations are checked to make sure the highlighted areas actually match what a real diagnosis would focus on. The goal isn't to replace doctors — it's to give health workers in under-resourced areas a helpful, honest second opinion, so patients get faster, safer care and serious cases don't get missed just because a specialist wasn't available.

Datasets

Dataset	URL link
ChestX-ray14 (NIH)	https://nihcc.app.box.com/v/ChestXray-NIHCC
Indiana University Chest X-Ray Collection (OpenI)	https://openi.nlm.nih.gov/faq#collection
ROCO Medical Dataset	https://github.com/razorx89/roco-dataset

Approach



Project Group Members

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